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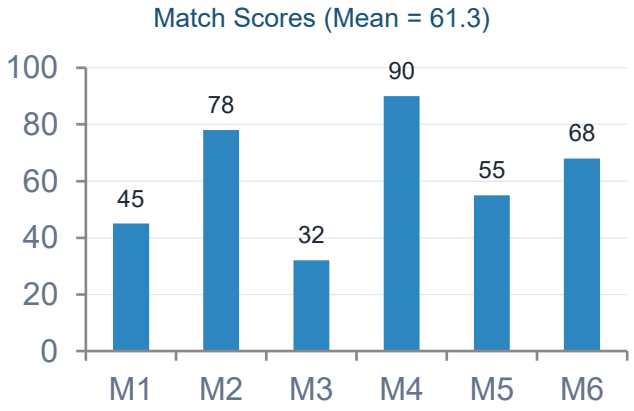
Statistics For Data Analysis

Mean (Arithmetic Average)

Mean is the average value of a dataset.

🏏 Real-World Example: Cricket Scores

Scores in 6 matches: **45, 78, 32, 90, 55, 68**
Mean = 61.3 runs



Key Points

- Uses all values in the dataset
- Gives the overall average
- Affected by extreme outliers
- Best for symmetric data

📊 Used In:
Salary Analysis · Temperature Avg
Exam Score Avg · Stock Price Avg

Median (Middle Value)

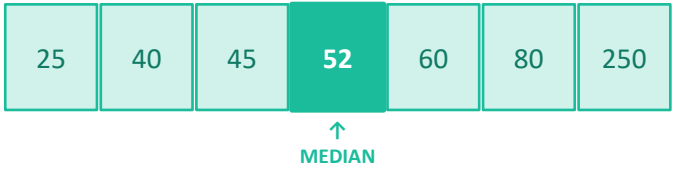
Median is the middle value in a dataset after arranging the values in ascending or descending order.

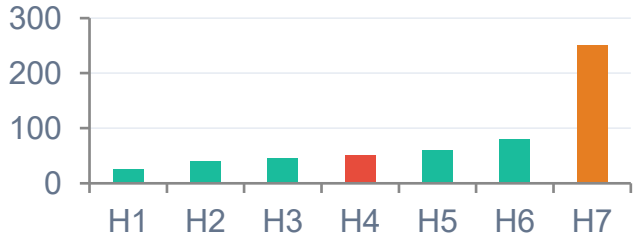
🏠 Real-World Example: House Prices (₹ Lakhs)

Sorted prices: **25, 40, 45, [52], 60, 80, 250**

Mean = 79.1 (skewed by ₹250L!)

Median = ₹52L → *More accurate!*





Key Points

- Middle value of ordered data
- Not affected by outliers
- Best for skewed data
- Divides data into two equal halves

📊 **Used In:**
House Price Analysis - Income Analysis - - Skewed Salary Data

Mode (Most Frequent Value)

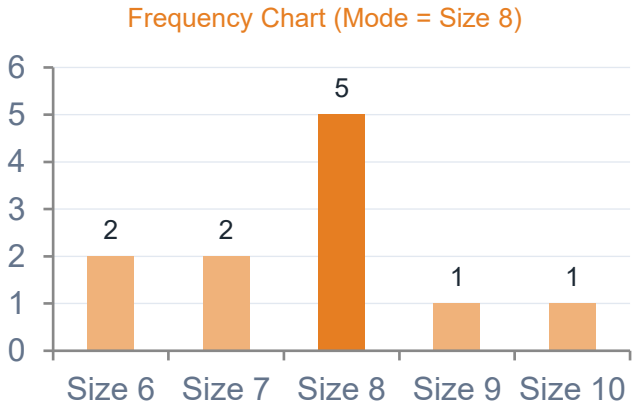
Mode is the value that appears most frequently in a dataset. A dataset can have one mode, more than one mode, or no mode.

👠 Real-World Example: Shoe Sales

Sizes sold: **6, 7, 8, 8, 9, 8, 7, 6, 8, 10, 8**

Mode = Size 8 (appears 5 times)

💡 *Stock more of size 8!*



Key Points

- Most frequently occurring value
- Useful for categorical data
- Can have multiple modes
- Shows the most common pattern/value

📊 **Used In:**
Most Sold Product - Most Common Shirt Size - Customer Preference Analysis

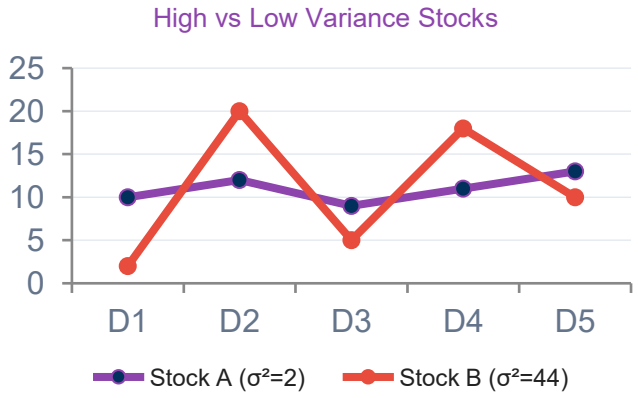
Variance (Spread of Data)

How the data values are spread out from the mean (average). It helps us understand the consistency or variability of the data.

📈 Real-World Example: Stock Returns (%)

Stock A: 10,12,9,11,13 → Mean=11
Variance = 2.0 → Low Risk ✓

Stock B: 2,20,5,18,10 → Mean=11
Variance = 44.0 → High Risk ✗



Key Points

- Data values are spread from the mean
- Higher variance means data is more spread out
- Lower variance means data is closer to the mean
- Calculated using squared differences from the mean

📊 Used In:
Risk Analysis - Stock Market Volatility - Exam Score Consistency - Quality Control Analysis

Standard Deviation (σ or s)

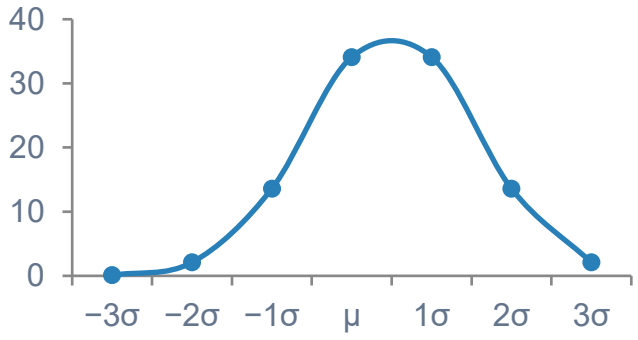
How much the data values are spread out from the mean (average).

🎓 Real-World Example: Exam Scores

Scores: 60, 65, 70, 75, 80 | Mean = 70

Variance = 50 → SD = √50 ≈ **7.07 marks**

68% students scored 62.9 – 77.1 (within 1 SD)



Empirical Rule (68–95–99.7)

- **μ ± 1σ = 68% of data**
- **μ ± 2σ = 95% of data**
- **μ ± 3σ = 99.7% of data**

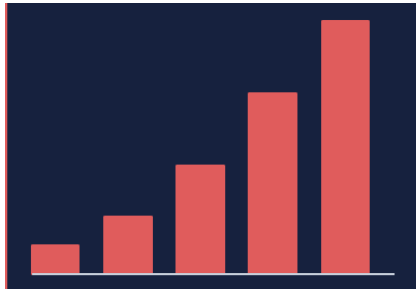
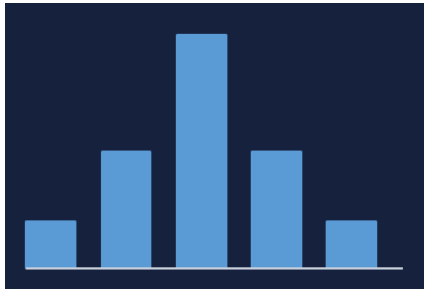
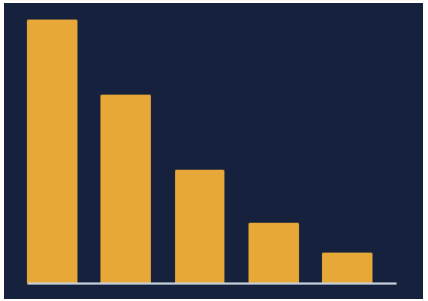
Key Points

- Measures the spread of data from the mean
- Low SD means data is close to the mean
- High SD means data is widely spread

📊 **Used In:**
Stock Market Risk Analysis - Student Performance Analysis - Weather Forecast Analysis

Skewness (Asymmetry of Distribution)

How the data distribution is symmetric or tilted toward one side. It helps identify the direction and extent of data imbalance.

Negative Skew	Symmetric	Positive Skew
		
Left-skewed Mean < Median < Mode Ex: Income tax payers	No skew Mean = Median = Mode Ex: Human height	Right-skewed Mode < Median < Mean Ex: Wealth distribution

Skewness Distribution

Skewness > 0 → Right tail is longer (Positive Skew)

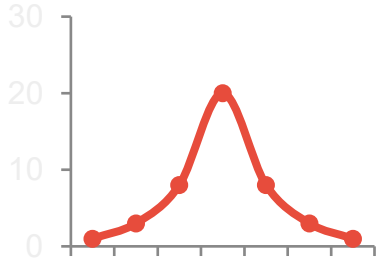
Skewness < 0 → Left tail is longer (Negative Skew)

Skewness = 0 → Data is symmetric (Normal Distribution)

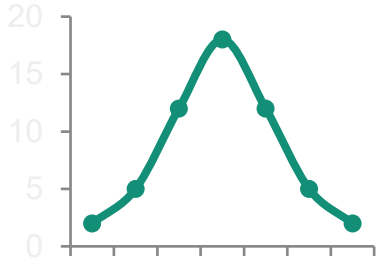
Kurtosis (Peakedness of Distribution)

How peaked or flat a data distribution is compared to a normal distribution. It helps identify the presence of extreme values or outliers in the data.

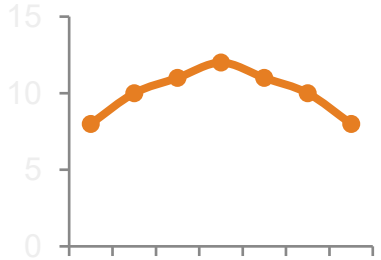
Leptokurtic	Mesokurtic	Platykurtic



Kurt > 3
Sharp peak, fat tails
Ex: Stock market crashes



Kurt = 3
Normal distribution
Ex: Human height



Kurt < 3
Flat peak, thin tails
Ex: Uniform outcomes

Kurtosis Distribution Classification
Kurtosis > 3 → Sharp peak & heavy tails (Leptokurtic)
Kurtosis = 3 → Normal peak distribution (Mesokurtic)
Kurtosis < 3 → Flat peak & light tails (Platykurtic)

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Key point of Statistics for Data Analyst

- Statistics helps convert raw data into meaningful insights
- Mean, Median, and Mode measure the central tendency of data
- Variance and Standard Deviation measure data spread and consistency
- Skewness identifies the symmetry and direction of a distribution
- Kurtosis identifies peakness and the presence of outliers
- Statistical analysis supports better decision-making
- Widely used in Business, Finance, Healthcare, and Data Analytics
- Helps identify trends, patterns, risks, and anomalies in data

💡 Together, these 7 measures fully describe the shape, center, spread, and symmetry of any dataset and help Decision Making!